

XC-algebras and quantum knot invariants

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Congreso Bienal de la RSME
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Slides available at bit.ly/jbecerra

Based on...

 Jorge Becerra Garrido.
Universal quantum knot invariants.
PhD thesis, University of Groningen, 2024.

 Jorge Becerra.
On Bar-Natan–van der Veen’s perturbed Gaussians.
Rev. R. Acad. Cienc. Exactas Fís. Nat. Ser. A Mat. RACSAM, 118(2):Paper No. 46, 58, 2024.

 Jorge Becerra.
A refined functorial universal tangle invariant.
arXiv:2501.17668.

 Jorge Becerra.
XC-tangles and universal invariants.
arXiv:2511.08045.

 Jorge Becerra and Kevin van Helden.
Minimal generating sets of rotational Reidemeister moves.
arXiv:2506.15628.

What are *quantum* knot invariants?

These are knot invariants whose construction is not intrinsic to the knot, but rather they have the extra data of some algebraic structure (a certain category, a TQFT, an algebra, a power series...). In some cases they admit extensions to quantum invariants of 3-manifolds.

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Some examples:

- The Jones polynomial,
- Khovanov homology,
- Khovanov-Rozansky / Lee homology,
- The Kontsevich invariant,
- $\vdots$

Today: the universal invariant

Let (H, R, v) be a *ribbon Hopf algebra*: this is a Hopf algebra $(H, \mu, \eta, \Delta, \varepsilon, S)$ over some ring $\mathbb{k}$ with

- a *universal R-matrix* $R \in H \otimes H$, i.e. an invertible element with

$$(\Delta \otimes \text{Id})R = R_{13} \cdot R_{23} \quad , \quad (\text{Id} \otimes \Delta)R = R_{13} \cdot R_{12} \quad , \quad \Delta^{op} = R \cdot \Delta(-) \cdot R^{-1}$$

- a *ribbon element* $v \in H$,

$$v \in \mathcal{Z}(A) \quad , \quad v^2 = uS(u) \quad , \quad \Delta(v) = (R_{21}R)^{-1}(v \otimes v) \quad , \quad \varepsilon(v) = 1 \quad , \quad S(v) = v$$

with $u = \mu^{op}(\text{Id} \otimes S)(R)$ the Drinfeld element. The ribbon element is automatically invertible.

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Theorem (Lawrence, Lee, Ohtsuki, Habiro,...)

Given a ribbon Hopf algebra (H, R, v) , one can obtain an invariant $\mathfrak{Z}_H(K) \in H$ of framed, oriented (long) knots.

The construction

Write

$$R = \sum_i \alpha_i \otimes \beta_i \quad , \quad R^{-1} = \bar{\alpha}_i \otimes \bar{\beta}_i \quad , \quad \kappa := uv^{-1}.$$

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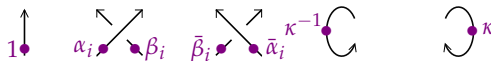
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Decorate these pieces with copies of $R^{\pm 1}$ and $\kappa^{\pm 1}$ as follows,

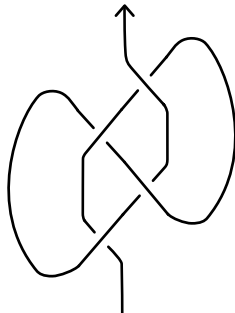


and then define $\mathfrak{Z}_H(K)$ as the element resulting from multiplying these beads from right to left following the orientation of the knot.

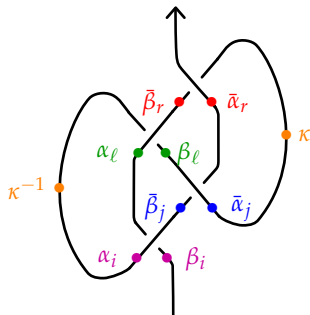
Example



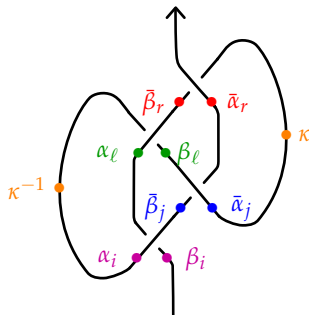
Example



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Example



$$3_H(4_1) = \sum_{i,j,\ell,r} \bar{\alpha}_r \bar{\beta}_j \alpha_i \kappa^{-1} \beta_\ell \bar{\alpha}_j \kappa \bar{\beta}_r \alpha_\ell \beta_i \in H$$

Why *universal*?

For any finite-dimensional representation (V, ρ) of H , the map

$$\rho(\mathfrak{Z}_H(K)) : V \longrightarrow V$$

equals the celebrated *Reshetikhin-Turaev invariant* $RT_V(K)$ obtained from the ribbon category $H\text{-mod}$ of finite-dimensional H -modules.

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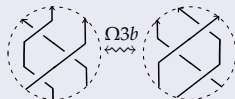
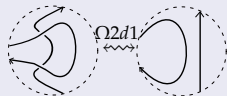
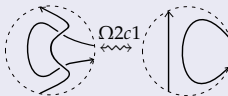
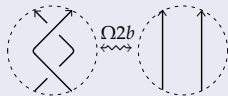
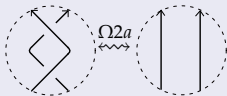
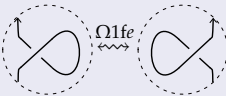
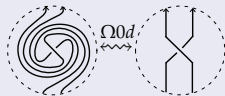
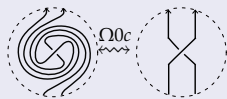
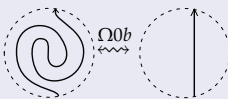
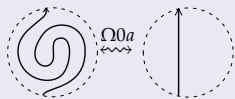
Key observation

The comultiplication, counit and the antipode of H are not used at all to construct $\mathfrak{Z}_H$!

Actually, to construct a knot invariant from this construction, we only need an algebra A with a couple of elements R, κ such that the Reidemeister moves are preserved!

Theorem (B.-van Helden 2025)

The following is a (non-minimal) generating set of rotational Reidemeister moves for rotational knot diagrams:



Definition

Let A be a $\mathbb{k}$ -algebra. An *XC-structure* on A is the choice of two invertible elements

$$R \in A \otimes A \quad , \quad \kappa \in A$$

satisfying

$$(XC0) \quad R^{\pm 1} = (\kappa \otimes \kappa) \cdot R^{\pm 1} \cdot (\kappa^{-1} \otimes \kappa^{-1}),$$

$$(XC1f) \quad \sum_i \beta_i \kappa \alpha_i = \sum_i \alpha_i \kappa^{-1} \beta_i ,$$

$$(XC2c) \quad 1 \otimes \kappa^{-1} = \sum_{i,j} \alpha_i \bar{\alpha}_j \otimes \bar{\beta}_j \kappa^{-1} \beta_i,$$

$$(XC2d) \quad \kappa \otimes 1 = \sum_{i,j} \bar{\alpha}_i \kappa \alpha_j \otimes \beta_j \bar{\beta}_i,$$

$$(XC3) \quad R_{12} R_{13} R_{23} = R_{23} R_{13} R_{12}.$$

The triple (A, R, κ) is called an *XC-algebra*.

An XC-algebra is the minimum algebraic structure needed to define a framed, oriented knot invariant.

Proposition

If (A, R, κ) is an XC-algebra, then the same construction of $\mathfrak{Z}_A(K) \in A$ from above gives rise to a well-defined knot invariant.

A few examples of XC-algebras

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3. There are XC-structures that do not have a ribbon Hopf-algebraic origin:
on the Sweedler algebra $SW = \langle s, w | s^2 = 1, w^2 = 0 \rangle$

$$R := 1 \otimes 1 + (1 + s + w + sw) \otimes (s + w + sw) \quad , \quad \kappa := -s - w - sw.$$

We have

$$\mathfrak{Z}_{SW}(\text{crossing}) = -1 - 2(s + w + sw) \notin \mathcal{Z}(SW) = \mathbb{k}1.$$

A few honest examples

Set $\mathbb{k} := \mathbb{Z}[q, q^{-1}]$ and $A := \text{End}_{\mathbb{k}}(\mathbb{k}^2) \cong \mathcal{M}_2(\mathbb{k})$.

Proposition

The elements

$$\begin{aligned} R := \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \otimes \begin{pmatrix} q & 0 \\ 0 & q^{-1} \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \otimes \begin{pmatrix} q^{-1} & 0 \\ 0 & q \end{pmatrix} \\ + (q - q^{-3}) \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \otimes \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \in A^{\otimes 2} \end{aligned}$$

and

$$\kappa := \begin{pmatrix} q^2 & 0 \\ 0 & q^{-2} \end{pmatrix} \in A$$

define a (traced) XC-structure on A .

Furthermore, for a 0-framed knot K we have that

$$\mathfrak{Z}_A(K) = J_2(K)|_{q^2 = -t^{-1/2}} \cdot \text{Id}_{\mathbb{k}^2}.$$

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Furthermore, for a 0-framed knot K we have that

$$3_A(K) = \Delta(K)|_{q^2=t^{1/2}} \cdot \text{Id}_{\mathbb{k}^2}.$$

The *Dilbert algebra* is

$$DLB = \langle d, l, b \mid dl = d, db = 1 - l, lb = b, l^2 = l = bd, \text{ others} = 0 \rangle.$$

Proposition

The elements

$$R := 1 \otimes 1 - 2(1 - l) \otimes l + 2b \otimes d \quad , \quad \kappa := \mathbf{i}(1 - 2l)$$

*define a (traced) XC-algebra structure on DLB.
Furthermore, for a 0-framed knot K we have that*

$$\mathfrak{Z}_{DLB}(K) = \Delta(K)(-1).$$

Theorem (B., to appear hopefully next week)

Any XC-algebra structure on the Sweedler algebra SW produces a framed knot invariant that only depends on the framing.

In particular, such an invariant is trivial for any 0-framed knot:

$$\mathfrak{Z}_{SW}(K) = 1.$$

Categorical framework

Most of the constructions in quantum topology are categorical/functorial (TQFTs, RT invariant, lasagna skein modules, Kh/Lee homology,...). The universal invariant did not have one even in the ribbon Hopf algebra setting.

Categorical framework

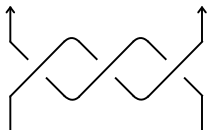
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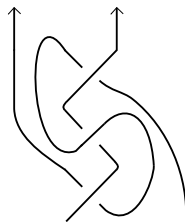
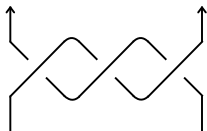
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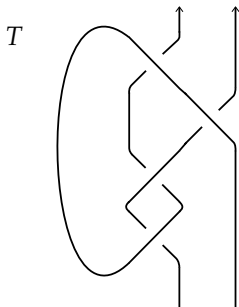
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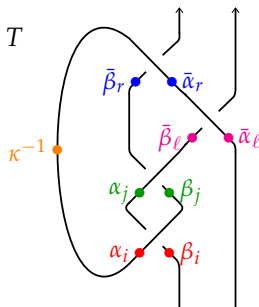
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$$\mathfrak{Z}_A(T) = \sum_{i,j,\ell,r} \bar{\beta}_\ell \alpha_j \beta_i \otimes \bar{\beta}_r \beta_j \alpha_i \kappa^{-1} \bar{\alpha}_r \bar{\alpha}_\ell \in A \otimes A,$$

Theorem (B. 2024)

Let (A, R, κ) be an XC-algebra. There exists a monoidal category $\mathcal{E}(A)$ and a strict monoidal full functor

$$Z_A : \mathcal{T}^{\text{up}} \longrightarrow \mathcal{E}(A)$$

which encodes the universal invariant $\mathfrak{Z}_A$:

$$Z_A(T) = (\mathfrak{Z}_A(T), \sigma_T).$$

Furthermore, this functor in fact arises in a canonical way – from a universal property.

If A is equipped with a trace, then one can in fact extend the construction to a functor

$$Z_A : \mathcal{T}^+ \longrightarrow \mathcal{E}^{\text{tr}}(A)$$

encoding $\mathfrak{Z}_A$ also arising canonically.

This setting allows to a functorial comparison with the Reshetikhin-Turaev invariant $RT_V : \mathcal{T}^{\text{up}} \longrightarrow H\text{-mod}$.

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Theorem (B. 2024)

Let H be a ribbon Hopf algebra and let V be a finite-free H -module. Then the Reshetikhin-Turaev invariant RT_V factors through Z_H :

$$\begin{array}{ccc} \mathcal{T}^{\text{up}} & \xrightarrow{RT_V} & H\text{-mod} \\ Z_H \downarrow & \nearrow \rho_V & \\ \mathcal{E}(H) & & \end{array}$$

That is, $RT_V(T) = \rho_V(Z_H(T))$ for any upwards tangle T . In other words, this diagram categorifies the equality

$$RT_V(K) = \rho(Z_H(K))$$

seen before.

If H is ribbon and V an H -module, we can produce two functors:

$$RT_V : \mathcal{T}^+ \longrightarrow H\text{-mod} \quad , \quad Z_{\text{End}_{\mathbb{k}}(V)} : \mathcal{T}^+ \longrightarrow \mathcal{E}(\text{End}_{\mathbb{k}}(V)).$$

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$$\begin{array}{ccc} & \mathcal{T}^+ & \\ Z_{\text{End}_{\mathbb{k}}(V)} \swarrow & & \searrow RT_V \\ \mathcal{E}(\text{End}_{\mathbb{k}}(V)) & \overset{\iota_V}{\dashrightarrow} & H\text{-mod} \end{array}$$

with ι_V a monoidal embedding. That is, viewing $\mathcal{E}(\text{End}(V))$ as a traced monoidal subcategory of $H\text{-mod}$, the functors $Z_{\text{End}_{\mathbb{k}}(V)}$ and RT_V coincide.

Extension to virtual tangles

A virtual knot is a knot diagram with positive, negative and *virtual* crossings –the latter are really not there!– modulo appropriate Reidemeister moves.

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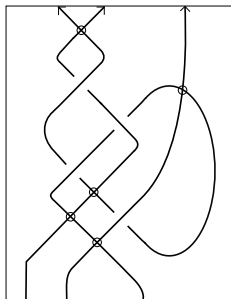
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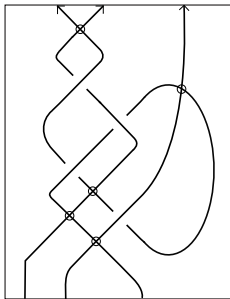
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Virtual knot = Gauss diagram on $\uparrow$
Virtual upwards tangle = Gauss diagram on $\Pi_n \uparrow$.

Theorem (B. 2025)

If A is an XC-algebra, then there exists a monoidal category $v\mathcal{E}(A)$ and a functor

$$Z_A : v\mathcal{T}^{\text{up}} \longrightarrow v\mathcal{E}(A)$$

extending the universal invariant of upwards tangles,

$$\begin{array}{ccc} \mathcal{T}^{\text{up}} & \xrightarrow{Z_A} & \mathcal{E}(A) \\ \downarrow & & \downarrow \\ v\mathcal{T}^{\text{up}} & \xrightarrow{Z_A} & v\mathcal{E}(A) \end{array}$$

More naturally, Z_A extends to the category $\mathcal{T}^{\text{XC}}$ of XC-tangles, a class of decorated abstract graphs that consists of the exact geometrical counterpart of XC-algebras.

Combining the last two theorems:

Corollary

Let H be a ribbon Hopf algebra and let V be a finite-free H -module. Then the invariant

$$Z_{\text{End}_{\mathbb{k}}(V)} : v\mathcal{T}^{\text{up}} \longrightarrow v\mathcal{E}(\text{End}_{\mathbb{k}}(V))$$

extends the Reshetikhin-Turaev invariant $RT_V : \mathcal{T}^{\text{up}} \longrightarrow H\text{-mod}$ to virtual upwards tangles.

¡Gracias por su atención!

Thank you – Dank u wel – Merci